



Neighbor feature variance (NFV) based feature point selection method for three dimensional (3D) registration of space target

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ABSTRACT

Point clouds based three dimensional (3D) registration for the space target pose estimation requires high precision and noise robustness. To enhance the registration accuracy, a novel neighbor feature variance (NFV) based feature points selection method is proposed to provide the high precision input of point clouds based registration and promote the pose estimation accuracy of space targets. The proposed method generates more accurate correspondences by selecting the rank of all point's NFV, for removing the redundant points and remaining more salient points for accurate registration. In coarse registration, the truncated least squares estimation and semi-definite relaxation (TEASER) algorithm is used to improve the accuracy of coarse registration and reduce the adverse effects caused by sparsity and noise of point cloud. In fine registration, the iterative closest point (ICP) algorithm is applied to estimate the pose transformation. The experimental results show that the maximum translation error and maximum rotation error are less than 0.019 m and 0.129°, and the mean translation and rotation errors can be reduced by 69.85% and 60.58%, respectively. Compared to the registration without the optimization algorithm, the proposed method can be used for variable scale and high noise points based registration.

1. Introduction

In-orbit technology provides in-orbit maintenance services for satellites, space stations, space vehicles and other space targets that fail, run out of fuel or need to replace modules [1]. Space targets can be classified into cooperative and non-cooperative targets based on whether detectable markers are installed beforehand [2]. A cooperative target is a target spacecraft fitted with a laser angle reflector or optical marker, or capable of actively communicating with the service spacecraft. At present, most spacecraft in orbit belong to non-cooperative targets, which cannot communicate with service spacecraft, and their structure, size and motion information are completely or partially unknown [3]. Relative pose measurement of non-cooperative targets is the premise and key point for the sustainable development of space rendezvous and docking [4,5].

The position and pose measurement for space targets mainly uses camera as sensor, which has been developed decades and its solving method is mature. The most common method is to extract the feature points in the image and then obtain the target pose by solving the perspective-and-point (PnP) problem [6–8]. However, when

illumination or target pose changes rapidly, the image is prone to overexposure, blur, distortion and other problems, resulting in inaccurate feature positioning or even unable to extract, which greatly affects the accuracy of pose estimation. With the blossom of three-dimensional sensors, such as light detection and ranging (LiDAR) and time-of-flight (TOF) camera, the pose measurement technology based on point cloud is gradually developed [9,10]. Compared to RGB images, the three-dimensional point cloud data of the target includes spatial coordinates and intensity information. Because of the characteristics of active detection, it is almost unaffected by light condition, which is suitable for measurement of the space targets. Since, three-dimensional point cloud data contains rich information and has unique advantages in the pose estimation of space targets.

At present, 3D point cloud has been applied to offer accurate relative pose for the unmanned rendezvous and docking of non-cooperative targets [11,12]. In 2006, the Japanese aerospace exploration agency [13] used binocular cameras for stereo matching of left and right images to obtain the target three-dimensional point cloud, and used iterative closest point (ICP) algorithm with the model point cloud to estimate the target pose. The three-axis translation error was less than 1 cm, and the

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rotation error was less than 5° . In 2012, the triangulation and LIDAR (TriDAR) system developed by Canadian Neptec company contained LiDAR and camera [14,15]. ICP algorithm was used to register the point cloud with the spacecraft model and complete the relative pose measurement, which was used for autonomous rendezvous, docking and relative navigation of non-cooperative targets. In 2016, Liu et al [16] used Flash LiDAR to acquire the target point cloud, and then used ICP algorithm to complete position estimation and pose tracking. In the simulation experiment, the translation error was less than 4 cm, and the rotation error was about 1° . In 2019, Li et al [17] proposed a state measurement method based on continuous point cloud, using extended Kalman filter to estimate the pose parameters of the target. A semi-physical experiment is carried out with the position error less than 5 cm, and the attitude angle error less than 3° . In 2020, Li et al [18] carried out coarse registration by extracting the satellite solar panel point cloud obtained by TOF camera, and then carried out ICP fine registration, so as to realize the position and pose tracking of non-cooperative satellites with rotation error less than 0.1° and translation error less than 2.5 mm. It can be seen that ICP method is used in most of the existing space target attitude estimation applications, which is the main method of point cloud fine registration [19]. However, the ICP method has high requirements on the initial value, which is easy to fall into the local optimal solution, resulting in registration failure, and the registration accuracy and efficiency are low, which is difficult to meet the registration requirements of massive point cloud data [20]. Therefore, the coarse registration method is needed to obtain the approximate matching relationship between two frames of point clouds before the fine registration of point clouds, so as to provide initial input for the fine registration.

The coarse registration methods mainly include random sample consensus (RANSAC) and principal component analysis (PCA) [21]. RANSAC method was proposed by Fischler and Bolles [22], and its basic principle is to traverse the whole point cloud with the idea of enumeration, determine the corresponding point relationship, and find the transformation matrix that minimization the objective function. Therefore, the RANSAC algorithm has a large time complexity and is not suitable for large-scale point cloud registration. Based on the principal direction, the PCA algorithm calculates the corresponding rotations and translations according to the principal direction of the point cloud, which can roughly align the two frames of point cloud data [23]. When the PCA method encounters the target with geometric structure, it is easy to have 180° angle deviation, which leads to coarse registration failure. Aiming at the impact of resolution and noise standard on point cloud registration algorithms, Yang et al. proposed a truncated least squares estimation and semidefinite relaxation (TEASER) global registration algorithm in 2021 [24]. It could solve the registration problem with a large number of external points, making the pose estimation results insensitive to mismatching. In this paper, the TEASER method is applied to the point cloud registration of space target before ICP, which can adapt to the point cloud with different resolution and noise standard and provide an initial input for fine registration.

In addition to the problem of point cloud quality, the geometric symmetry of space targets causes more mismatching pairs, and it is difficult to deal with point clouds with large data scale due to limited space computing resources. Before point cloud registration, voxel filtering method or key point extraction is usually used to obtain small volume point cloud. The relatively robust ISS method for key point extraction was proposed by Zhong in 2009 [25,26]. This algorithm extracts key points by calculating the eigenvalues of the covariance matrix of the query points using the neighborhood information. The disadvantage of ISS is that the characteristic points obtained by ISS are concentrated. If a point is judged as a key point, its neighborhood points are highly likely to be judged as key points because they have similar neighborhood information. Therefore, lots of key points obtained by ISS method are distributing over a close region and relatively concentrated, which cannot represent the feature points diversity of overall target. In

order to improve the accuracy of point cloud registration and reduce the calculation amount, a point cloud selection method based on point cloud neighbor feature variance (NFV) is proposed to remove a large number of redundant point clouds. After the key points selection, TEASER coarse registration and ICP fine registration are used to achieve the point cloud registration of space targets, which provides high-precision data basis for the attitude estimation of space targets.

The remaining of this paper is organized as follows: Section 2 describes the point selection and registration method and the evaluation indicators. A series of simulation are performed to verify the proposed method on the space target point cloud with different resolution and noise standard in Section 3. Results demonstrate the effectiveness and robustness of the proposed method. Conclusions are presented in Section 4. Section 5 has discussed the implications of the findings and future research direction.

2. Methodology

Difficulties in the registration of satellite target point cloud is that the high inverse material of satellite target causes more noise points, and the geometric symmetry of satellite target causes more mismatched pairs. Moreover, it is difficult to deal with the point cloud with large data scale due to limited spatial computing resources. In order to solve the above problems, a point cloud selection method based on neighbor feature variance (NFV) is proposed in this paper, and truncated least squares estimation and semidefinite relaxation (TEASER) combined with iterative closest point (ICP) is used for point cloud matching, so as to realize the satellite target point cloud registration with high robustness. The flowchart of the methodology is shown in Fig. 1 and the logical diagram is shown in Fig. 2.

2.1. Point cloud acquisition with varying quality

Point clouds of satellite targets are obtained by LiDAR or other three-dimensional sensors, and their quality is related to the sensor's own performance and scanning factors. The data quality of point cloud can be evaluated from two aspects: point cloud resolution and point cloud noise standard. The point cloud resolution is divided into model point cloud resolution (mpr) and sampled point cloud resolution (spr), and is the average of the nearest neighbor distance between model point cloud and sampled point cloud. Model point cloud resolution can be calculated as:

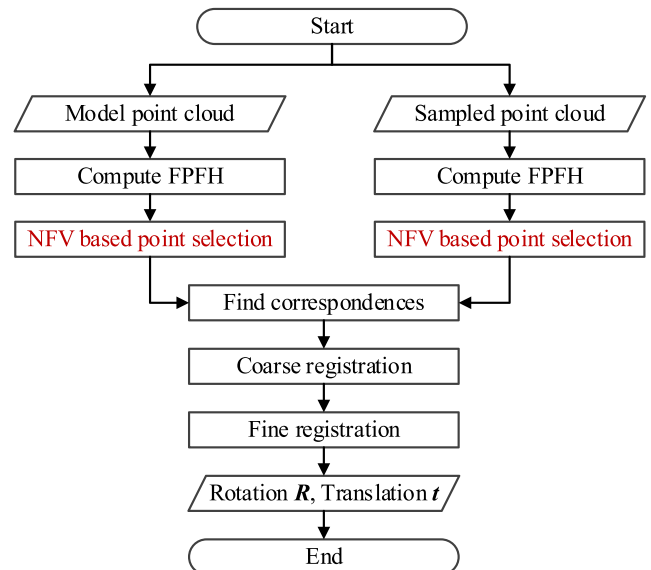


Fig. 1. Flowchart of the methodology.

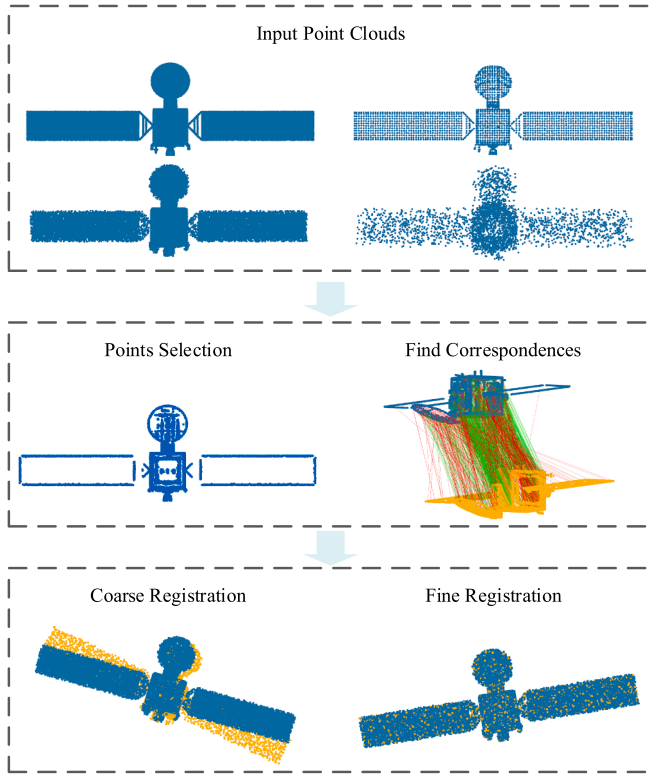


Fig. 2. Logical diagram of the methodology.

$$\text{mpr} = \frac{1}{n} \sum_{i=1}^n d_i \quad (1)$$

where n represents the number of points in the model point cloud, and d_i represents the distance between each point and its nearest neighbour.

Sampled point cloud is obtained by down-sampling of model point cloud, so the resolution of sampled point cloud can be expressed as multiples of model point cloud resolution:

$$\text{spr} = s_{\text{sampled}} \cdot \text{mpr} \quad (2)$$

where s_{sampled} is the sampled multiple of the sampled point cloud.

There are two main types of point noise, one is caused by the deviation between the measurement and the real values, and the other is caused by system noise disturbance or background strong light. The noise of the former is still belong to the target point cloud, while the latter is generally regarded as the outlier of the target points. The outliers can be removed by setting the radius filter in the noise filtering process, which will not be discussed in this paper. The noise standard discussed in this paper is mainly aimed at the system noise of points cloud, which can be simulated based on superimposing Gaussian noise over the truth value. The point cloud noise standard is defined as the root mean square error of the noise of the point cloud. Point cloud noise can be expressed as the multiples of the resolution of the sampled point cloud:

$$\sigma_{\text{noise}} = s_{\text{noise}} \cdot \text{SPF} \quad (3)$$

where, s_{noise} is the noise multiple factor of point cloud.

2.2. Neighbor feature variance-based point selection

Because the points close each others are redundant computation for registration, so it is necessary to remove lots of redundant points to enhance the registration accuracy. This step aims to select a subset of points from point cloud representing the target, because the raw point

cloud is usually quite large and requires high computational complexity, resulting in unbearable computing time. Meanwhile, depending on the sensors and target, too much data also brings redundant or unnecessary details into the task of registration. Therefore, point selection as a pre-processing step to reduce the number of points is imperative in registration with high computational efficiency and accuracy. Many down-sampling strategies have been proposed and studied for point selection, such as random, distance limit, and uniform down-sampling. However, these methods simply apply points selection indiscriminately to each part of the target. We propose neighbor feature variance-based point selection method for elimination of redundancy.

First of all, fast point feature histograms (FPFH) descriptor is used to describe the neighborhood geometry information of a point in the point cloud [27]. As a kind of point cloud feature descriptor, FPFH has rotation and scale invariance, high fault tolerance, and strong robustness to point cloud data containing noise. The FPFH descriptor for point $\mathbf{p}$ can be noted as $\text{FPFH}(\mathbf{p})$. A FPFH descriptor at a point $\mathbf{p}$ relies on the presence of 3D coordinate and estimated surface normals, and is computed as follows: i) for each point $\mathbf{p}$, all of $\mathbf{p}$'s neighbors enclosed in the sphere with a given radius r are selected (k -neighborhood); ii) for every pair of points $\mathbf{p}$ and $\mathbf{p}_i$ in the k -neighborhood of $\mathbf{p}$ and their estimated normals $\mathbf{n}$ and $\mathbf{n}_i$, a Darboux $\mathbf{u}\mathbf{v}\mathbf{w}$ frame is defined ($\mathbf{u} = \mathbf{n}$, $\mathbf{v} = (\mathbf{p} - \mathbf{p}_i) \times \mathbf{u}$, $\mathbf{w} = \mathbf{u} \times \mathbf{v}$) and the angular variations of $\mathbf{n}$ and $\mathbf{n}_i$ (Fig. 3) are computed as follows:

$$\begin{cases} \alpha = \mathbf{v} \cdot \mathbf{n}_i \\ \phi = \frac{\mathbf{u} \cdot (\mathbf{p} - \mathbf{p}_i)}{\|\mathbf{p} - \mathbf{p}_i\|} \\ \theta = \arctan(\mathbf{w} \cdot \mathbf{n}_i, \mathbf{u} \cdot \mathbf{n}_i) \end{cases} \quad (4)$$

The relationship of angular variations between $\mathbf{p}$ and its neighbors is called Simplified Point Feature Histogram (SPFH). For each point $\mathbf{p}$, we re-determine its k neighbors and use the neighboring SPFH values to weight the final histogram of $\mathbf{p}$ (called FPFH):

$$\text{FPFH}(\mathbf{p}) = \text{SPFH}(\mathbf{p}) + \frac{1}{k} \sum_{i=1}^k \frac{1}{w_i} \cdot \text{SPFH}(\mathbf{p}_i) \quad (5)$$

Next, the concept of neighbor feature variance is proposed to quantify the significance for each feature descriptor (see Fig. 4). The neighbor feature variance measures the difference between the feature descriptor of a point with those of the points within the neighborhood, which can be calculated as:

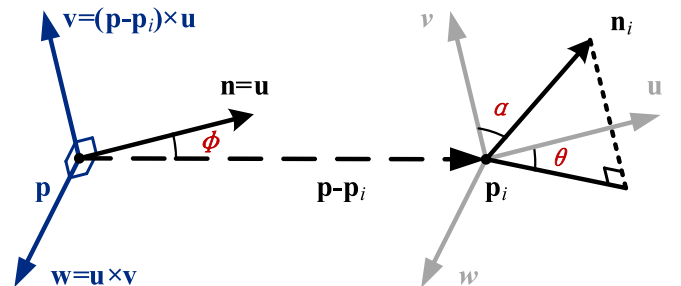
$$\sigma^2(\mathbf{p}) = \left| \text{FPFH}(\mathbf{p}) - \frac{1}{k} \sum_{i=1}^k \text{FPFH}(\mathbf{p}_i) \right| \quad (6)$$

where, k the number of the points within the neighbourhood.

In order to normalize NFV, which is expressed as $\hat{\sigma}^2(\mathbf{p})$:

$$\hat{\sigma}^2(\mathbf{p}) = \sigma^2(\mathbf{p}) / \max(\sigma^2) \quad (7)$$

where, $\hat{\sigma}^2(\mathbf{p})$ represents the normalized neighbor feature variance. The greater the $\hat{\sigma}^2(\mathbf{p})$, the greater the difference between the calculated

Fig. 3. The angular variations of point $\mathbf{p}$ and its neighbor point $\mathbf{p}_i$.

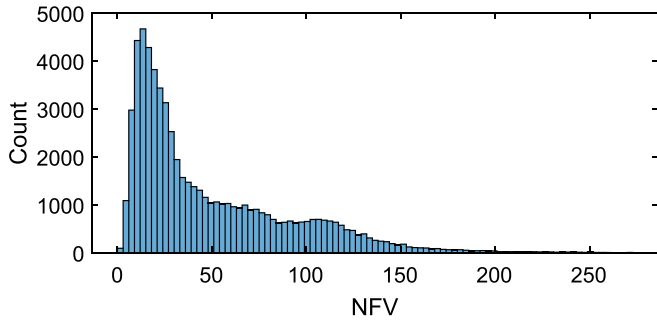


Fig. 4. The histograms of neighbor feature variance for point cloud.

point and the neighborhood points, and the greater important role the calculated point plays in the point cloud.

Finally, according to the neighbor feature variance, the points can be ranked as critical. The principle of the neighbor feature variance screening method based on descriptor proposed in this paper is that the points are arranged in descending order according to the normalized neighbor feature variance, and the top points are retained as the key points. The NFV value quantifies the feature difference between a point and its neighborhood points. The rule of point cloud selection is to first set the threshold of NFV as σ_{th}^2 , then remove points whose NFV are smaller than the threshold, and finally achieve key point extraction. As shown in the histograms of NFV values of satellite targets, it can be seen that NFV values of a large number of point clouds are concentrated in a small area, these points are non-convex points distributing over plane of satellite, and points greater than the threshold of NFV are convex points and can be used for better registration computation.

Take $\mathbf{p}_r$ (point in red circle) and $\mathbf{p}_b$ (point in blue square) as convex point and non-convex point. The FPFH vectors of $\mathbf{p}_r$ and $\mathbf{p}_b$ and their neighborhood points are shown in Fig. 5. $\mathbf{p}_r$ is located in the main body of the satellite, and the FPFH vector of $\mathbf{p}_r$ is very different from that of the neighboring points, so its NFV value is also large with $\sigma^2(\mathbf{p}_r) = 0.9980$. $\mathbf{p}_b$ is located in the solar panel, and the difference between the FPFH vector with that of the neighborhood points is small, indicating that it is a non-convex point with $\sigma^2(\mathbf{p}_b) = 0.0599$. The NFV value is able to quantify the feature difference between a point and its neighborhood points. By setting the threshold of NFV, points whose NFV smaller than the threshold can be removed, thus achieving key point extraction.

2.3. Brief review of the registration algorithm

The registration method adopted in this paper includes three steps: matching pair formation, coarse registration and fine registration. The matching pair is formed to provide matching correspondence for coarse registration. Coarse registration calculates the best transformation matrix of the point cloud based on the matching pairs, and fine registration optimizes the transformation matrix based on the coordinate positions of points.

Find correspondences: The purpose of correspondences formation is to discover the pairs of every point in the sampled point cloud onto the model point cloud so that the correspondences are established between the two sets of point clouds. We can find the correct translation and rotation between the two sets of point clouds from the appropriate correspondences. In this paper, based on the calculated features, each point in the sampled point cloud is traversed to search the matching points in the model point cloud, and the nearest neighbor in the model point cloud is found to form a matching pair.

Coarse registration: A new algorithm called truncated least squares estimation and semidefinite relaxation (TEASER) is used for coarse registration, and traditional Random sample consensus (RANSAC) method is used as a contrast. TEASER algorithm is a global registration algorithm proposed by Yang in 2021. It remodels the registration

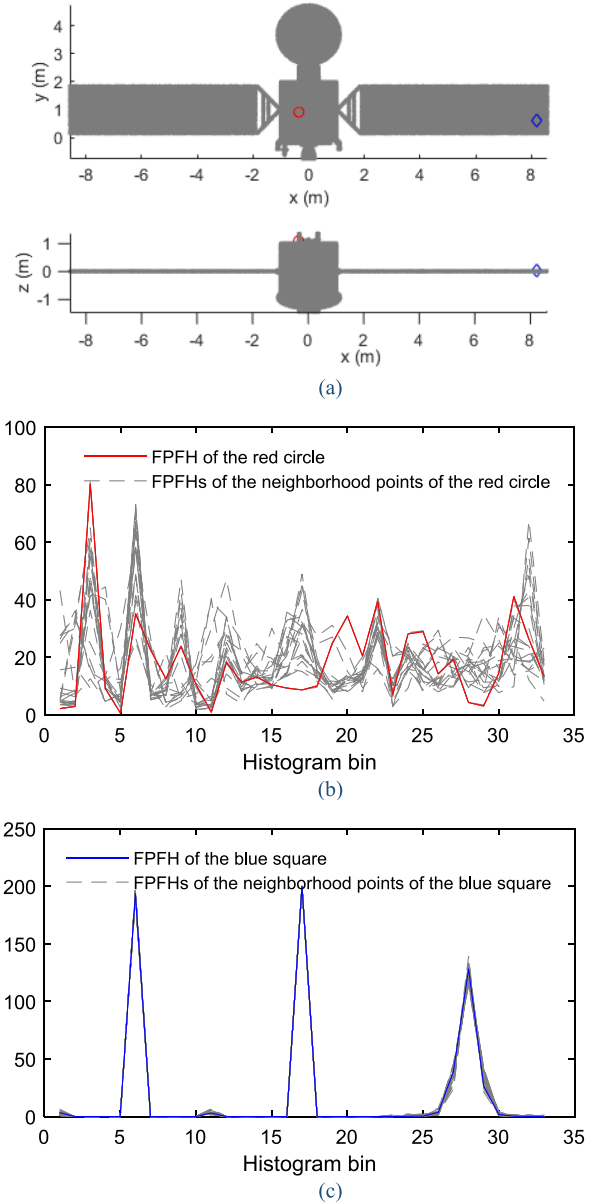


Fig. 5. The FPFH vectors of point cloud, (a) point cloud with $\mathbf{p}_r$ and $\mathbf{p}_b$; (b) FPFH vectors of $\mathbf{p}_r$ and its neighborhood points; (c) FPFH vectors of $\mathbf{p}_b$ and its neighborhood points.

problem and uses semidefinite relaxation to solve the nonlinear and non-convex optimization problem. It can robustly solve the registration problem with a large number of outliers, making the pose estimation results insensitive to mismatching. TEASER uses graduated non-convexity to solve the rotation subproblem and leverages Douglas-Rachford Splitting to efficiently certify global optimality. A framework is provided to decouple scale, translation and rotation transformation, and the estimation can be expressed as:

$$\min_{s>0, \mathbf{R} \in SO(3), \mathbf{t} \in \mathbb{R}^3} \sum_{i=1}^n \frac{1}{\sigma_i^2} \|\mathbf{b}_i - s\mathbf{R}\mathbf{a}_i - \mathbf{t}\|^2 \quad (8)$$

where s , $\mathbf{R}$ and $\mathbf{t}$ are the unknown scale, rotation and translation, σ_i models the measurement noise.

In practice, a large fraction of the correspondences are outliers due to incorrect keypoint matching. Single “invalid” outlier can compromise the correctness of the resulting estimate. Hence, truncated least squares registration formulation is proposed that can tolerate extreme amounts

of spurious data.

$$\min_{s>0, R \in SO(3), t \in \mathbb{R}^3} \min_{i=1}^n \left(\frac{1}{\beta_i^2} \|b_i - sRa_i - t\|^2, \quad \bar{c}^2 \right) \quad (9)$$

where, β_i is a given bound and $\bar{c}$ is a constant.

RANSAC method is a typical coarse registration algorithm based on exhaustive search, and its core idea is randomness and hypothesis. Randomness is to randomly select the sampling data according to the probability of the correct data. According to the law of large numbers, the randomness simulation can approximate the correct result. Hypothesis is to assume that the selected sampling data are correct, and then use these data to solve the problem, and count the number of inliers of the calculation result. Through several iterations, the model with the largest number of internal points is taken as the final result.

Fine registration: Iterative closest point (ICP) is the most classical fine registration method in point cloud registration, which has attracted extensive attention since it was proposed. The basic idea is to find out the corresponding point in the target point cloud for each point in the source point cloud according to the constraints, and then iteratively calculate the transformation matrix until the error function is minimized.

2.4. Evaluation indicators for registration

In order to quantitatively evaluate the performance of point cloud registration, different evaluation indicators are proposed for the three stages of point cloud selection, matching pair formation, point cloud registration.

In terms of point cloud section, repeatability is proposed to describe the ability to accurately detect the same key points under various disturbances, and defined as the corresponding percentage of the key points detected in two set of point clouds after being transferred to the same coordinate system, which can be calculated as Rep:

$$\text{Rep} = \frac{N_{\cap}}{N_{\cup}} \times 100\% \quad (10)$$

where, $N_{\cap}$ represents the number of the intersection of two sets of point clouds and $N_{\cup}$ the number of the union of two sets of point clouds.

In the stage of matching pair formation, inlier ratio (IR) is applied to measure the fraction of correct matches in the predicted correspondences set M_{pred} . A match is correct if it lies within a distance threshold d_{th} after the transformation using the ground truth transformation matrix T_{gt} . IR is defined as:

$$\text{IR} = \frac{1}{|M_{\text{pred}}|} \sum_{(p,q) \in M_{\text{pred}}} [\|T_{\text{gt}}p - q\|_2 < d_{\text{th}}] \quad (11)$$

where, $\|\cdot\|_2$ is the Euclidean norm and $[\cdot]$ is the Iverson bracket. (p, q) is the match member in the predicted correspondences set M_{pred} .

In addition, matching point deviation (MPD) is proposed to evaluate the quality of the predicted matching pairs. For each matching pair, due to the limited sample rate of the sensor and the existence of noise and outliers, there is a deviation distance between the two points in the found matching pairs. MDP is defined as the sum deviation distance between each two points in the predicted correspondences set M_{pred} .

In terms of point cloud registration, to evaluate the difference between the predicted transformation and the ground truth transformation, we adapt rotation error and translation error which are commonly used for the evaluation of point cloud registration. For a given target point cloud Q , the predicted transformation matrix T_{pred} from source point cloud P can be calculated by registration algorithms. The ground truth transformation matrix between two sets of point clouds is given as T_{gt} . The residual transformation ΔT from Q to P is defined as Eq. (12):

$$\Delta T = T_{\text{pred}}(T_{\text{gt}})^{-1} = \begin{bmatrix} \Delta R & \Delta t \\ \mathbf{0} & 1 \end{bmatrix} \quad (12)$$

where, ΔR is the residual rotation matrix in 3 by 3 dimensional, Δt is the residual translation vector in 3 by 1 dimensional.

Then, we can define the error of rotation e_{rot} and the error of translation e_{tra} as:

$$\begin{cases} e_{\text{rot}} = \arccos\left(\frac{\text{trace}(\Delta R) - 1}{2}\right) \cdot \frac{180}{\pi} \\ e_{\text{tra}} = \|\Delta t\|_2 \end{cases} \quad (13)$$

where $\text{trace}(\Delta R)$ represents the trace of residual rotation matrix (ΔR), and $\|\Delta t\|_2$ is the 2-norm of the residual translation vector Δt .

3. Simulation and results

In order to verify the reliability of the neighbor feature variance-based point selection method, the following simulation is designed and intrinsic shape signatures (ISS) method is taken as a comparison method. First, the two methods are used to obtain different percentage points and offer a direct evaluation of the point selection results. Secondly, the robustness of the two methods to point clouds with different resolutions is estimated by repeatability. Finally, the accuracy of matching pairs between the selected point cloud and the model point cloud is studied by inlier ratio. Thus, the effectiveness of the NFV-based point selection method is demonstrated by the simulation results.

3.1. Overview for NFV of satellite target

The simulation uses the point cloud of Beidou satellite, which is $17.1541 \times 5.5064 \times 2.8129$ (m) in length, width and height, and contains a total of 83,261 points.

The satellite target is divided into four areas according to the geometric distribution. Area 1 is the main body of the satellite including the skeleton connected to the solar panel, area 2 is the circular antenna, area 3 is the docking ring and the tail nozzle, and area 4 contains the left and right solar panel. The neighbor feature variance of the above four areas is shown in the Fig. 6, among which the distribution range of area 1 is widest from 0.04 to 1.00. When the NFV is greater than 0.66, only points in area 1 remain, because there are more details with higher NFV in area 1, and only points in area 4 with neighbor feature variance less than 0.04, because area 4 has high feature homogeneity and contains many redundant points. The histogram distribution of characteristic variance in area 4 presents two peaks, in which the upper point of the peak at the front of the distribution is concentrated in the interior of the solar panel, and the lower point is concentrated in the edge of the solar panel.

According to the reserved points of neighbor feature variance, the ratio of the reserved points to the original points was defined as the retention rate, which was represented by reservation ratio (RR). The NFV histograms for four areas with varying reservation ratio is shown in Fig. 6 (b). Setting the appropriate RR can remove redundant points, such as when the RR is less than 50%, most of the points in area 4 have been removed. The point cloud of the satellite target is colored according to NFV as shown in Fig. 7. It can be seen that the NFV is small on the plane of the solar panel and the main body, while the neighbor feature variance is larger on the edge of the satellite and on the main body of the antenna, docking ring, nozzle and other typical parts. Therefore, the point selection based NFV can be used to eliminate redundant points.

3.2. NFV-based points selection compared to ISS

To verify the effectiveness of the proposed NFV point selection method, the conventional and common method intrinsic shape signatures (ISS) is used as a comparison method. ISS method was introduced

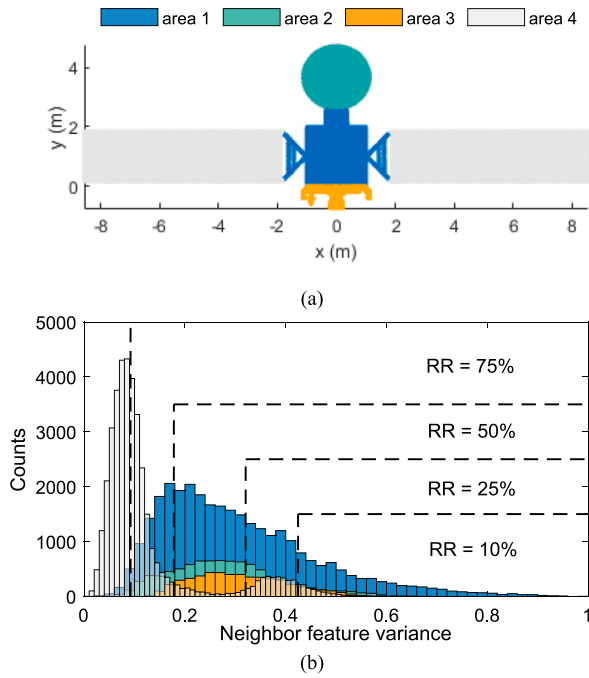


Fig. 6. NfV of the point cloud of satellite target (a) the satellite includes main body (area 1), antenna (area 2), docking ring and nozzle (area 3) and solar panel (area4); (b) the NfV histograms for four areas with varying reservation ratio.

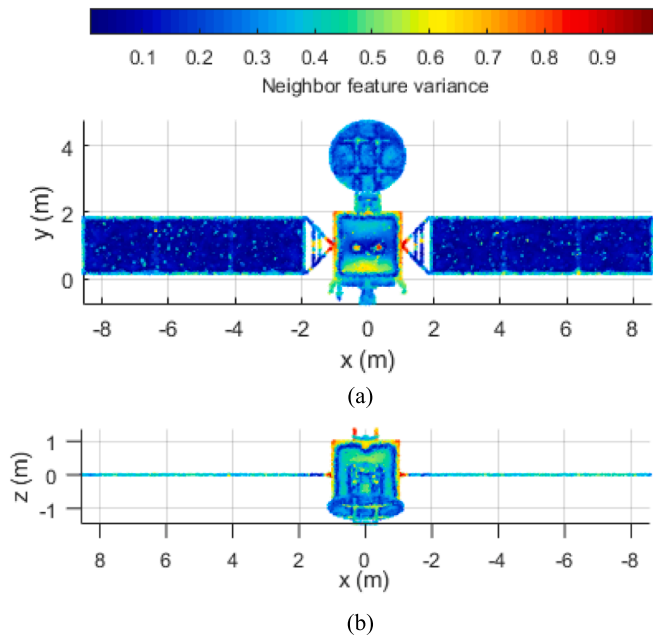


Fig. 7. Point cloud of the satellite target colored according to the NfV. (a) front view, (b) top view.

by Zhong based on the eigenvalue decomposition of the scatter matrix of the points, which was proved to be with the important advantage of being highly efficient and robustness. The following will compare the two methods from three aspects to prove the performance of the proposed NfV method in point cloud selection.

3.2.1. Comparison on point cloud selection results

The selected points are combined with feature descriptors to form feature point descriptors, which can form a compact representation of

the original data, thus increasing the processing speed of subsequent algorithms. Firstly, the NfV-based point selection method and the ISS method are used to obtain different percentage points and offer an intuitive evaluation of the point selection results.

The point cloud of the satellite target is retained of about 50% and 10% from the original points as shown in Fig. 8. By comparing the point cloud selection results of the two methods, it can be intuitively seen that the point cloud of the proposed method has a better retention effect on the contour of satellite targets. When the number of points is about 50% of the model point cloud, the point cloud obtained by ISS method retains solar panel contour points, satellite main body points and antenna contour points. The method proposed in this paper not only preserves the above feature points, but also characterizes the vertical rod of the solar panel and leaves the inner point of the antenna. When the point cloud is about 10% of the model point cloud, the point cloud obtained by ISS method is only the outline of the solar panel and antenna, and there is hardly any point of the main body. However, the method proposed in this paper preserves the satellite body, and these points will be available in point cloud registration. This is because in terms of the principles of the two methods, the NfV-based point selection method selects the points with larger feature differences from its neighborhood points, which is easy to retain some prominent parts. The points obtained by ISS method are more inclined to the points with prominent normal directions, such as the edge points of the plane. However, the feature points of the satellite are more concentrated on the main body of satellite target. Therefore, the NfV based point selection method has an advantage over the ISS method on point selection of satellite target.

Next, the evaluation indicator of repeatability is used to verify the performance of the point cloud selection method of different reservation ratio under varying point cloud conditions. When the resolution and noise standard of the point cloud remain unchanged, for the two sets of point clouds with rotation and translation transformation, the repeatability of the selected points by the two methods reaches 100%. The reason is that the NfV-based point selection method is based on FPFH descriptor and the ISS method is calculated on the premise of establishing local coordinate system, both of them have the characteristics of rotation and translation invariance. The results prove the robust of the proposed NfV-based point selection method against the rigid transformation.

In order to further verify the robustness of the proposed point selection method for point clouds under various conditions, the simulation is carried to display the repeatability of the selected point clouds. The resolution of the point cloud is changed from 2 mpr to 8 mpr with the noise standard of 1 spr and the repeatability results are shown in Fig. 9. It is obvious that the repeatability of the method proposed in the paper is higher than the ISS method under almost all point cloud conditions.

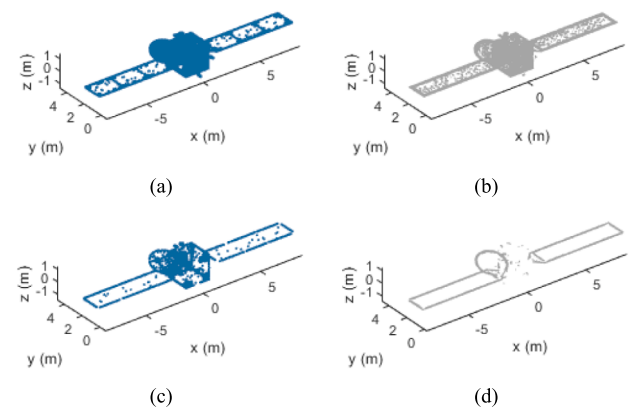


Fig. 8. Results of the selected points (a) 50.00% by the neighbor feature variance-based point selection method (b) 51.40% by the ISS method (c) 10.00% by the NfV method (d) 10.56% by the ISS method.

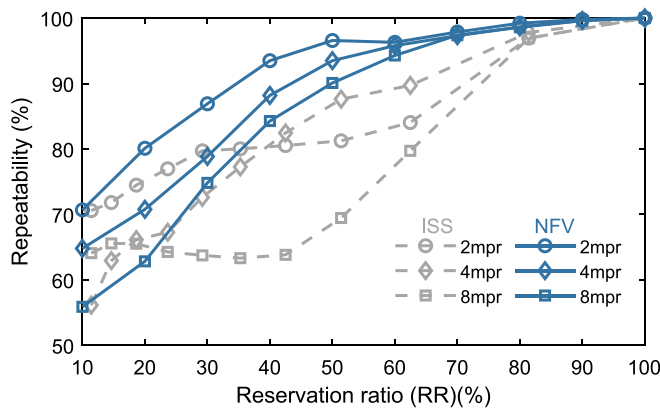


Fig. 9. The repeatability results of the point clouds with different resolutions (blue: the proposed method, grey: the ISS method). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Therefore, the proposed NFW point cloud selection method has a predominance in terms of robustness.

3.2.2. Comparison on quality of matching pairs

The purpose of point selection is to provide more concise point cloud data so as to complete efficient and accurate point cloud registration. The matching pairs obtained by the model point cloud and the sampled point cloud without point cloud selection are shown in Fig. 10 (a). Due to the symmetry property of the geometric structure of the satellite

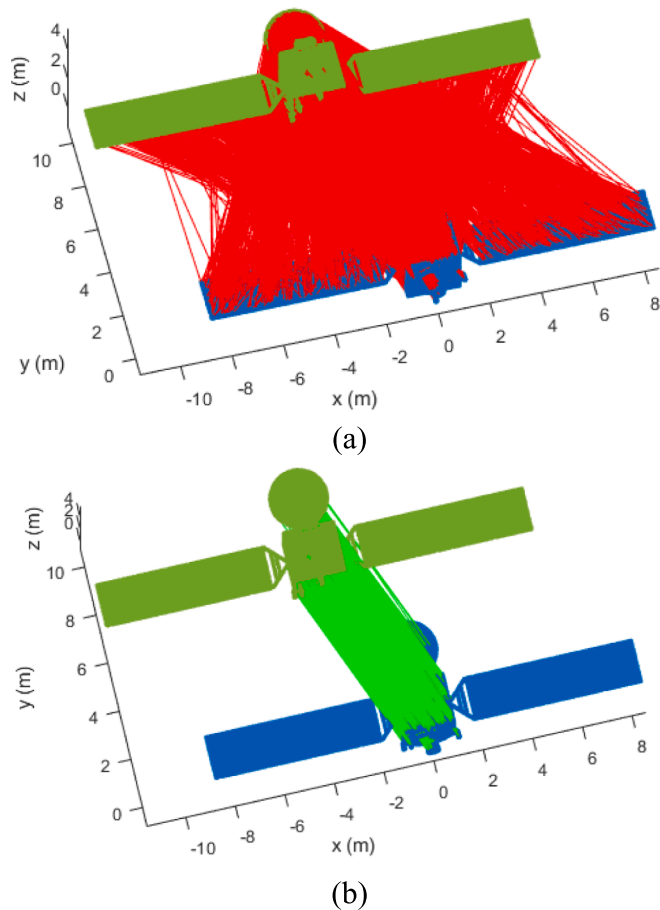


Fig. 10. The matching pairs between the model point cloud and the sampled point cloud (a) before points selection (b) after NFW-based points selection.

target, there are a large number of wrong matching pairs. The correct matching pairs are found less than one-fifth and almost concentrated in the main body part of the satellite target, as shown in Fig. 10 (b). Since the main points obtained after NFW point cloud selection are satellite main body points, it can be preliminarily estimated that NFW point cloud selection is beneficial to form correct matching pairs.

The simulation is aiming to verify the effectiveness and accuracy of the proposed NFW point cloud selection method. The effectiveness and accuracy of matching pairs between the selected point cloud and the model point cloud are studied by inlier ratio (IR) and matching point deviation (MPD). After selecting point clouds with different resolutions by two methods, matching pairs between them and the model point cloud are found based on FPFH descriptors (Fig. 11), and the inlier ratio and matching point deviation of matching pairs is calculated, as shown in Fig. 12 and Fig. 13. The IR ($d_{th}=3$ mpr) of matching pairs for the proposed NFW-based point selection method is better than that of the ISS method when the point cloud resolution decreases. The IR of matching pairs of the selected points by the proposed method is better than that of point cloud without selected (100%), which is contrast to the results by the ISS method. The IR of the point cloud selected by the NFW method increases 38.85% by the point cloud without selection. This is because the function of the method proposed in this paper is to filter redundant points with high feature similarity, so as to achieve the removal of preliminary mismatching pairs and improve the efficiency of subsequent point cloud registration.

In order to further prove the effectiveness of the matching pairs formed after point cloud selection, matching point deviation (MPD) is used as an index to evaluate the matching pairs obtained by the two

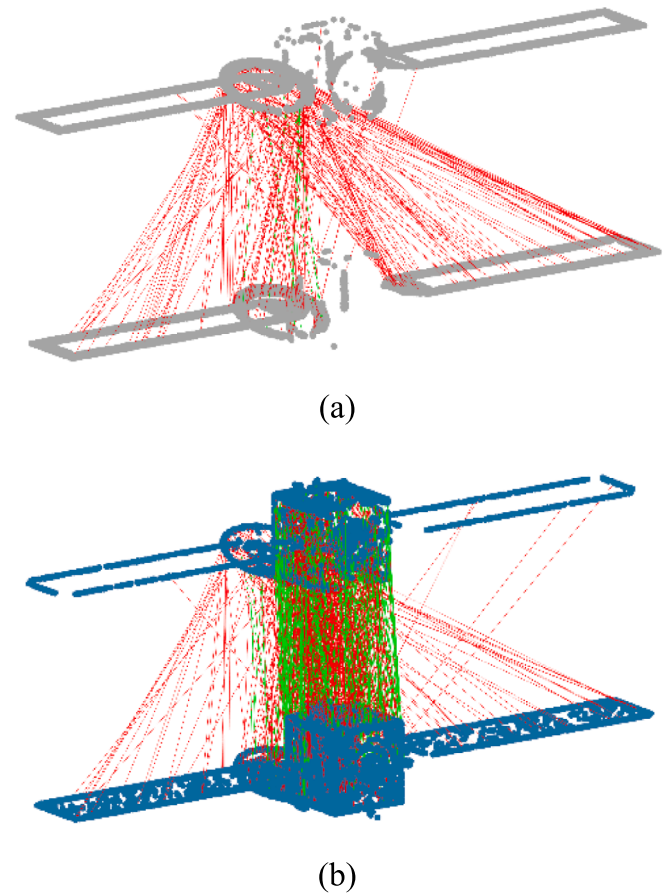


Fig. 11. The matching pairs of the selected point clouds with the wrong in red and true in green, (a) the ISS method, (b) the NFW method. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

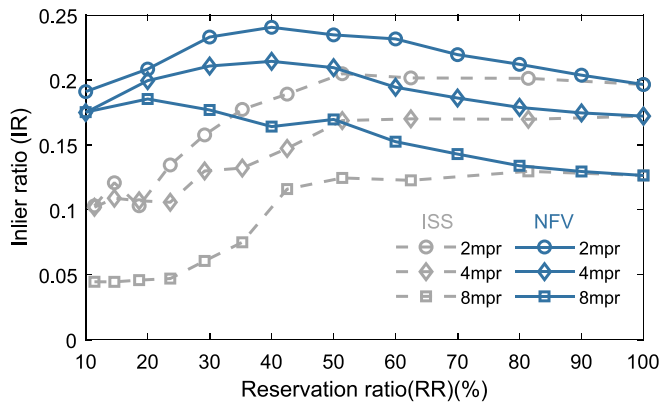


Fig. 12. The inlier ratio results of the point clouds with different resolutions (blue: the proposed method, grey: the ISS method). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

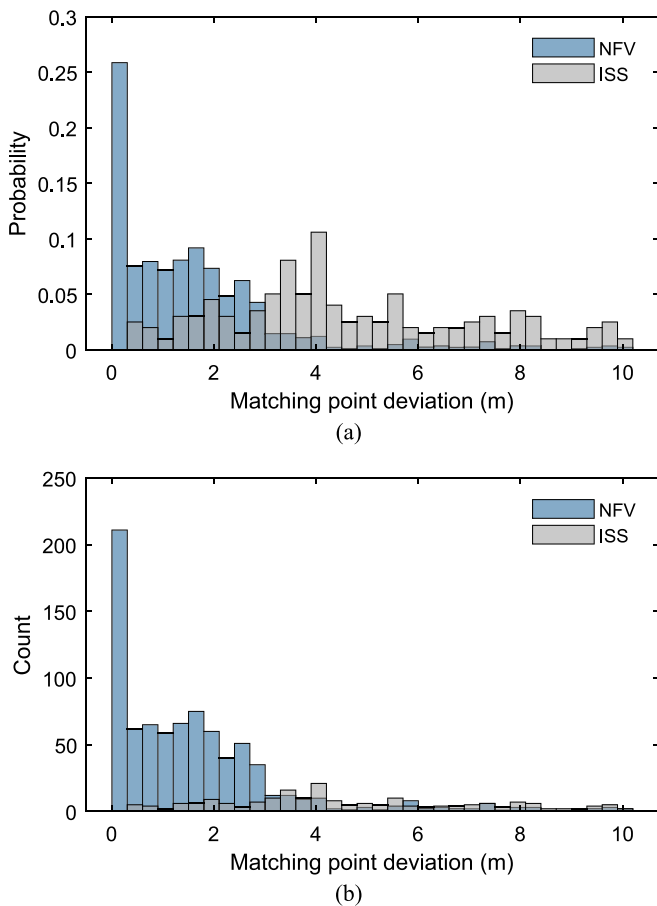


Fig. 13. The matching point deviation of the point clouds with different points selection method, (a) the probability of the MPD, (b) the count of the MPD (blue: the proposed method, grey: the ISS method). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

methods. Fig. 13 shows the probability histogram and count histogram of MPD for the NfV method and ISS method. Whether in terms of relative probability or absolute count, the MPD obtained by the NfV method is absolutely dominant in a relatively small area. It is proved that the NfV method produces more accurate matching pairs than the ISS method. High-quality selected points can improve the accuracy of

matching pairs, reduce mismatches, and improve the subsequent point cloud registration results.

3.3. NfV verification with point registration

The main goal of point cloud registration is to find the optimal transformation matrix between two different point clouds. Point cloud registration can be divided into coarse registration and fine registration. The experiment of this paper firstly compares the proposed point cloud registration methods in coarse registration and fine registration steps respectively, and finally verifies the advantages of the proposed point cloud selection method in point cloud registration. In this simulation experiment, we obtained the on-time cloud data to be deployed under different distances and perspectives of 20 types of satellite points cloud, as shown in Fig. 14. The point resolution is changed by down-sampling and the noise standard is changed by adding Gaussian white noise to simulate the change of point cloud quality caused by different system design.”.

3.3.1. Coarse registration by TEASER and RANSAC

The function of coarse point cloud registration is to roughly align two frames of point clouds that are far apart with the aim of providing a good initial estimate for subsequent fine registration. It is necessary to applying coarse registration before fine registration for two frame point clouds with large displacement and rotation angles. Aiming at the influence of resolution and noise standard on point cloud registration algorithm, truncated least squares estimation and semidefinite relaxation (TEASER) registration algorithm are used to improve the accuracy of coarse registration.

In order to compare the robustness of RANSAC and TEASER coarse registration algorithms to the change of point cloud resolution (1 mpr ~ 8 mpr) and noise standard (1 spr ~ 11 spr), this paper conducted a comparison experiment of coarse registration results, as shown in Fig. 15.

The resolution and noise standard of the point cloud are changed respectively, and RANSAC and TEASER algorithms are used to achieve coarse registration. The rotation error and translation error obtained are shown in Fig. 16. It can be concluded that the registration result of TEASER algorithm is better than RANSAC in both point cloud resolution and point cloud noise standard, and the error stability is better in low noise scenarios. This is because RANSAC is based on the idea of exhaustive search, which assumes that the random matching pairs selected each time are correct. However, in the actual practice, due to the similarity of the local structure of the target or the interference of noise points, mismatches will be produced, thus greatly affecting the solution results. Facing the robust solution of registration problem, TEASER uses truncated least square function as the cost function for a large number of external points in registration, to avoid the influence of mismatching on registration results to a certain extent. Therefore, TEASER algorithm is more robust than RANSAC algorithm in the face of mismatches caused by environmental noise or sparse point clouds.

3.3.2. Fine registration by ICP

In order to further optimize the pose estimation results, this paper apply the ICP fine registration to TEASER coarse registration results, and

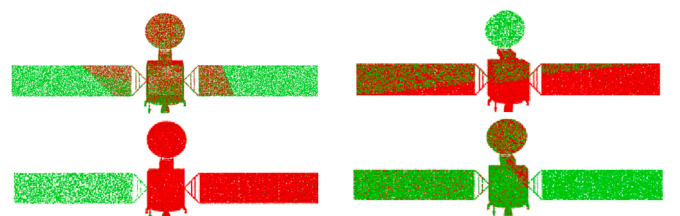


Fig. 14. The different conditions of the sampled point cloud.



Fig. 15. The registration results of (a) RANSAC (b) TEASER (blue is the model point cloud, yellow is the sampled point cloud after registration). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

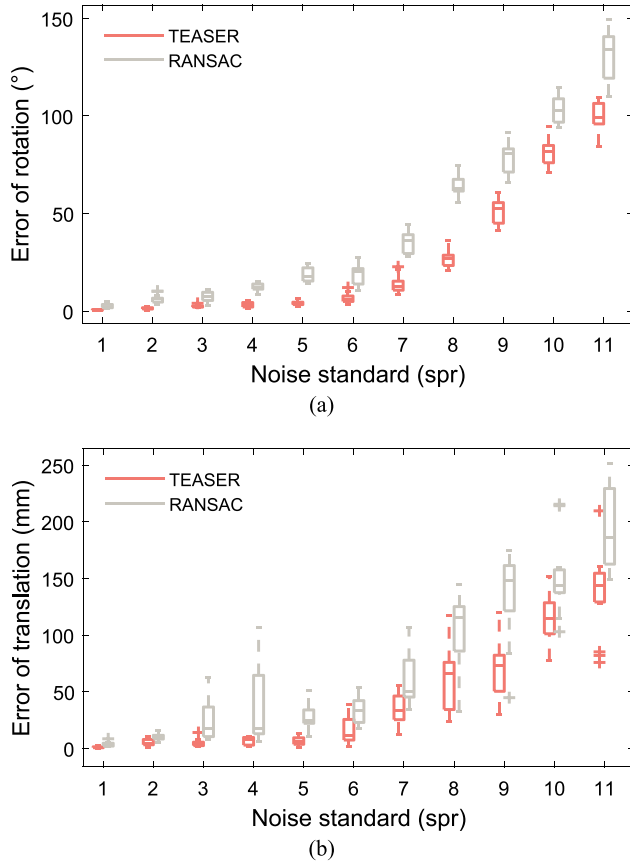


Fig. 16. The registration errors of TEASER and RANSAC method under different noise standard (a) rotation errors (b) translation errors.

finally get the exact pose estimation value of the target. The alignment results are shown in Fig. 17.

In order to verify the effectiveness of the TEASER combined with ICP method proposed in this paper, the registration results of the TEASER method and the combined with ICP method are compared. As is shown in Fig. 18, the rotation error decreased significantly, and the translation



Fig. 17. The registration result of TEASER combined ICP method (blue is the model point cloud, yellow is the sampled point cloud after registration). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

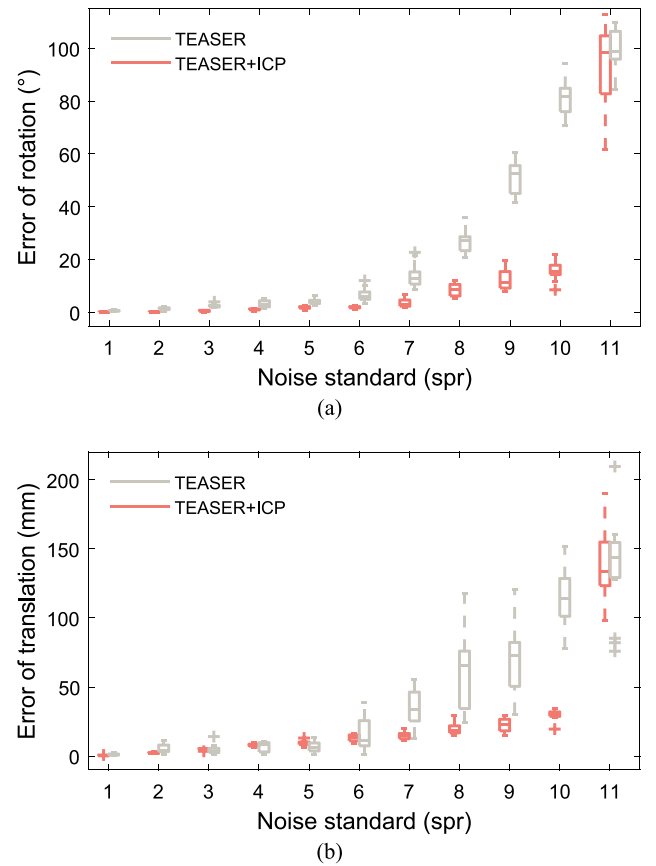


Fig. 18. The registration errors of TEASER and TEASER combined ICP method under different noise standard (a) rotation errors (b) translation errors.

error also kept in a low range of the ICP method. Specifically, the translation error is less than 1 mm and the rotation error is less than 0.5° when there is no noise. When the noise standard is less than 5 spr, the maximum translation error is 0.017 m and the maximum rotation error is 2.671° , which still has a high accuracy. When the noise continued to increase to 10 spr, ICP could not provide accurate initial pose after coarse registration due to the serious loss of structural information of point cloud itself.

3.3.3. Registration with NFV points selection

In the above experiments, the efficiency and robustness of the proposed point cloud registration method have been proved. In order to further confirm the important role that the NFV point cloud selection method plays in the point cloud registration, this part carries out a comparison experiment of registration errors between the selected points by NFV method and voxel down-sampling. In the experiment, the model point cloud of satellite target and the sampled point cloud with resolution of 2 mpr and noise standard of 2 spr are used as the data basis to be registered. The NFV method and voxel down-sampling method are applied to obtain the point cloud with RR of 10%~100%, and the translation and rotation error of point cloud registration are shown in Fig. 19. As can be seen from Fig. 19, the registration error using NFV point cloud selection almost does not increase with the decrease of the number of point clouds, the translation errors are less than 0.019 m, and the rotation errors less than 0.129° . The mean translation and rotation errors can be reduction by 69.85% and 60.58% respectively after the NFV based point selection method. This shows that the point cloud selection method proposed in this paper can achieve point cloud simplification and enhance the registration accuracy. Experimental results under different RR prove the robustness and effectiveness of the proposed method.

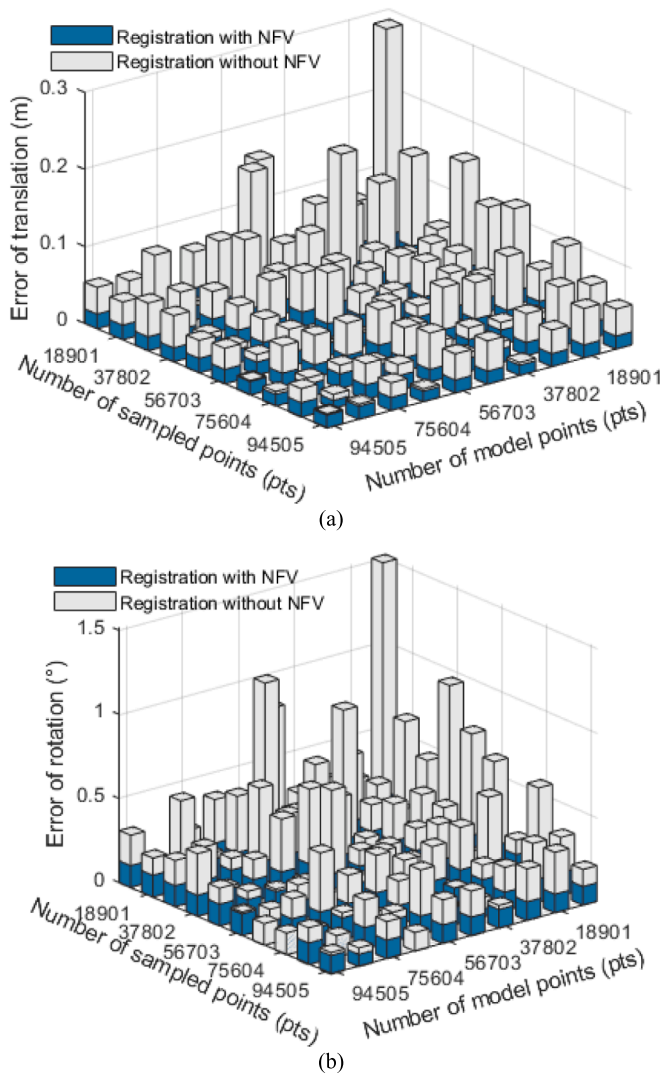


Fig. 19. The errors of registration with NfV and without NfV method under different number of points (a) rotation errors (b) translation errors. The registration algorithm includes TEASR and ICP.

4. Discussion

Through the proposed point cloud selection and registration method, the results show a wide range of applicability and robustness under different data quality. It can handle various noises and errors, and has high registration accuracy and reliability. The simulation results emphasized that, point resolution and noise standard are the dominant influencing parameters that change point cloud quality and affect point cloud registration performance. Through the evaluation of the point cloud registration experiments, three points were noted:

- (1) *Enhance the correspondence accuracy:* Compared to ISS and RAN-SAC, NfV effectively select the feature points to generate more accurate matching pairs, IR increases 38.85% by the point cloud without selection and registration accuracy improves by two to ten times.
- (2) *Improve registration robustness:* The noise standard increases to 5 spr and the resolution reaches 5 mpr as the proposed method achieves the registration under the centimeters and degrees of the translation and rotation error.
- (3) *Reduce time consumption:* The proposed method can complete 3D registration in only tens of milliseconds when the number of

correspondences is smaller than 1000. Note that registration is implemented on the CPU only. In this paper, the algorithm is implemented on the CPU, and the execution time can reach 600 ms/10000 points. The model of the CPU is i7-11700 K@3.60 GHz, with 16 GB of memory and 8 cores. CPU is mainly used for single-threaded computing and consumes less power than the GPU, making it more practical for future space missions.

These experimental results provide researchers a useful guide for space targets detection and identification in the future. The resolution and noise standard of point cloud should be determined to optimize the LiDAR systematic error range. However, the proposed method has the drawback of key points extraction accuracy limited by determination of neighborhood radius of NfV, which is mainly influenced by target sizes and point clouds resolutions. Seeking a reasonable way to realize the adaptive radius setting for different scale and resolution of point cloud will gain better insight into this subject in the future work.

5. Conclusion

To conquer the issues of the influence of target scale variation and noise interference on registration accuracy degradation for non-cooperative target, a neighbor feature variance (NfV) based feature point selection method for 3D registration is proposed to extract the feature points precisely and reliably for further coarse and fine registrations. The neighbor feature variance of each point can be calculated from feature descriptors correlation of the points and neighbor points within a certain neighbor area, determining more accurate feature correspondences by ranking the NfV as the weight for each point. The feature correspondences with few redundancies are used to high-precision estimate the pose transformation of non-cooperative target.

Based on the model data of Beidou satellite as a space target representation, a series of comprehensive experiments are conducted to demonstrate the effectiveness and performance of the proposed method for space target pose estimation. The experimental results show that, when the point noise standard is twice the point resolution, the maximum translation error and maximum rotation error for registration are less than 0.019 m and 0.129°. Compared with the registration without the proposed feature point selection method, the mean translation and rotation errors can be reduced by 69.85% and 60.58%, respectively. The NfV based feature point selection method for 3D registration is capable to address the challenges of variable scale and high noise points based registration. Other than the three-dimensional information of points used in this contribution, the intensity of the point cloud can be used as additional information to further enhance the registration performance of space noncooperative target in future work.

CRedit authorship contribution statement

Yier Zhou: Conceptualization, Methodology, Software, Writing – review & editing. **Xiaolu Li:** Data curation, Supervision. **Haixia Hu:** Investigation. **Lixuan Su:** Visualization. **Hang Du:** Software, Validation. **Wenming Fu:** Writing – review & editing. **Lijun Xu:** Supervision.

Declaration of Competing Interest

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Data availability

Data will be made available on request.

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